

# Equations, functions and graphs

**Solving equations, what a function is, and the shapes you have to recognise on sight.**

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- 1.1 Linear equations and systems
  - 1.2 Quadratic equations in business
  - 1.3 What a function is
  - 1.4 Domain and range
  - 1.5 Six families of function
  - 1.6 Exponential functions
  - 1.7 Logarithmic functions
  - 1.8 Week 1 review quiz
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## 1.1 Linear equations and systems

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- 
- ▶ **Video** First order equations · Why we do the same thing to both sides
  - ▶ **Video** Systems, by substitution · Solving systems of equations with substitution
  - ≡ **Article** The method, written out · Multi-step equations review
- 

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- ✍ **Exercise** First order equations · Equations with variables on both sides
  - ✍ **Exercise** Systems, by elimination · Systems of equations with elimination
- 

### FIRST ORDER

One unknown, first power.  $ax + b = 0$  with  $a \neq 0$ .

### A SYSTEM

Two equations that have to hold at the same time, for the same pair of values.

- 1 **Clear brackets, then fractions.** Multiply every term by the common denominator, the right-hand side included.
- 2 **Collect the unknown on one side, the numbers on the other.**
- 3 **Divide by the coefficient.**

$$ax + b = 0 \quad \Rightarrow \quad x = \frac{-b}{a}$$

- 4 **For a system, eliminate.** Scale one equation so a letter cancels when you add the two.
- 5 **Or substitute.** Make one equation say  $x = \text{something}$ , and put that into the *other* one.
- 6 **Check the pair in both equations.** One equation cannot catch an error.

### THREE OUTCOMES

One solution. None, when you reach something false. Infinitely many, when you reach something always true.

None of the three is a mistake.

## First order equations

$$x = \frac{-b}{a}$$

WORKED:  $3x + 1 = 5x + 7$

$$3x + 1 = 5x + 7$$

Collect the  $x$  terms on one side, the numbers on the other.

$$-2x = 6$$

$$3x - 5x = -2x \text{ and } 7 - 1 = 6.$$

$$x = -3$$

Divide by  $-2$ .

## Systems of first order equations

### THE SAME SYSTEM, BOTH WAYS

$$2x + y = 18 \text{ and } -x + 5y = 2$$

The system to solve.

$$-2x + 10y = 4$$

Method 1: multiply the second equation by 2, then add it to the first.

$$11y = 22, \text{ so } y = 2 \text{ and } x = 8$$

The x terms cancel.

$$x = 5y - 2$$

Method 2: isolate x in the second equation instead.

$$2(5y - 2) + y = 18$$

Substitute into the first.

$$11y = 22, \text{ so } y = 2 \text{ and } x = 8$$

The same answer, as it must be.

### WATCH OUT

- Multiplying an equation in a system means multiplying its right-hand side too.
- $-(x - 4) = -x + 4$ . Losing that second sign is the commonest slip here.
- Check your pair in both equations. One is not enough.

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## 1.2 Quadratic equations in business

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- ▶ **Video**     The formula · The quadratic formula
  - ▶ **Video**     What the discriminant tells you · Using the quadratic formula: number of solutions
  - ≡ **Article**    The discriminant · Discriminant review
- 

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- ✍ **Exercise**   Using the formula · Quadratic formula
  - ✍ **Exercise**   Reading the discriminant · Number of solutions of quadratic equations
- 

- 1    **Everything to one side.**

$$ax^2 + bx + c = 0, \quad a \neq 0$$

- 2    **Read off a, b and c with their signs.** The signs are where the marks go.
- 3    **Work out the discriminant first.**

$$\Delta = b^2 - 4ac \quad \Delta > 0: \text{two roots} \quad \Delta = 0: \text{one} \quad \Delta < 0: \text{none in } \mathbb{R}$$

- 4    **Factor if it factors, otherwise use the formula.**

$$x = \frac{-b \pm \sqrt{\Delta}}{2a}$$

- 5    **In a business question, break-even is where revenue meets cost.**  $R(q) = C(q)$  rearranges to a quadratic you already know how to solve.
- 6    **Throw away a root the situation forbids.** A negative quantity or a negative price is not an answer.

$$x_1 = \frac{-b + \sqrt{\Delta}}{2a} \quad x_2 = \frac{-b - \sqrt{\Delta}}{2a}$$

**WORKED:**  $3x^2 + 3x - 18 = 0$

$$\Delta = 3^2 - 4(3)(-18) = 225$$

Positive, so two solutions.

$$x = \frac{-3 \pm 15}{6}$$

$$\sqrt{225} = 15.$$

$$x = 2 \text{ or } x = -3$$

The two roots.

$$3x^2 + 3x - 18 = 3(x + 3)(x - 2)$$

It also factors directly, which is faster when the numbers are kind.

## The application: Berger Tools

**THE SOLUTION**

$$C(q) = 6q + 80\,000$$

Total cost: variable plus fixed.

$$R(q) = 10q$$

Total revenue: price times quantity.

$$P(q) = R(q) - C(q) = 4q - 80\,000$$

Profit is the difference of the two functions.

$$60\,000 = 4q - 80\,000$$

Set the profit to the target.

$$q = 35\,000 \text{ units}$$

Which is the answer to the business question, not just to the algebra.

**WATCH OUT**

- $\Delta < 0$  means no solution in  $\mathbb{R}$ . Write that; it is a complete answer.
- In  $-4ac$  with  $c$  negative the two signs make a plus. Most wrong discriminants come from here.
- Move everything to one side before reading off  $a$ ,  $b$  and  $c$ .

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## 1.3 What a function is

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- 
- ▶ **Video** What counts as a function · Recognizing functions
  - ▶ **Video** Evaluating a function · Worked example: Evaluating functions from equation
  - ▶ **Video** Lines, slope and intercept · Worked examples: slope-intercept intro
- 

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- 
-  **Exercise** Evaluating · Evaluate functions
  -  **Exercise** The vertical line test · Recognize functions from graphs
  -  **Exercise** Reading a line off a graph · Slope-intercept equation from graph
- 

### FUNCTION

A rule that gives each allowed input **exactly one** output. That is the whole definition.

### F(X)

The output at x. It is not f times x.

### FOUR REPRESENTATIONS

A formula, a table, a graph, a sentence. The same function, four ways, and you are expected to move between them.

### VERTICAL LINE TEST

If a vertical line meets the graph twice, it is not a function: that input would have two outputs.

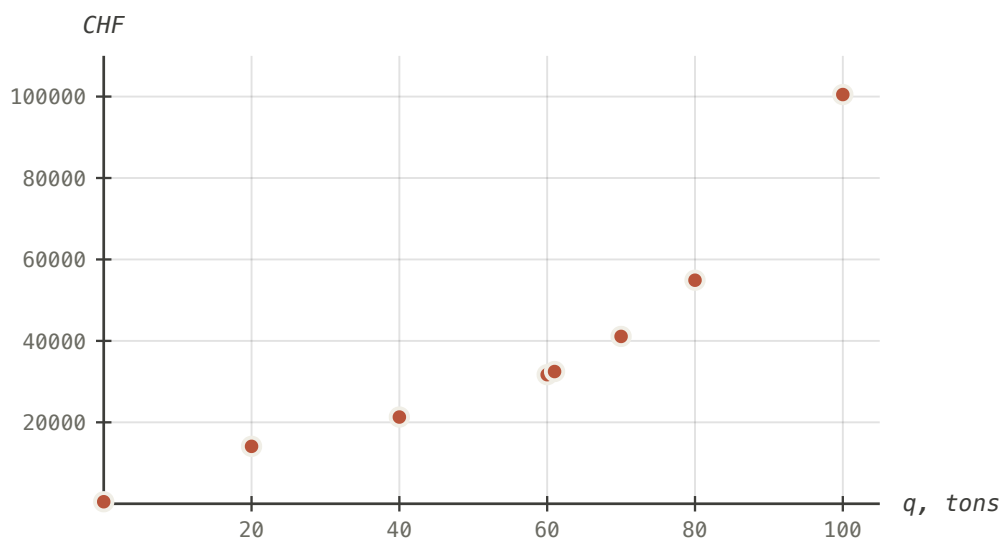
- 1 **To evaluate, replace every x by the whole input.** Brackets included:  $f(a + 1)$  needs  $(a + 1)$  everywhere.
- 2 **Read a table exactly, a graph approximately.** A graph gives you the shape and the trend, not the fourth decimal.
- 3 **Slope is rise over run, and it is the same wherever you measure it.**

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

- 4 To sketch a line, plot two points and join them. The intercepts are usually the two easiest.
- 5 A vertical line has no slope.  $x = k$  is not a function, and its slope is undefined rather than zero.

## The course definition

### The worked case: Rivella Coffee



The Rivella Coffee cost table, drawn. Cost rises with quantity, and faster at the top end.

## The four representations

EVALUATING, AND WHY BRACKETS MATTER

$$h(x) = 2$$

A constant function: every input gives 2.

$$h(10) = 2, h(-213) = 2, h(x + 3) = 2$$

The third one makes the point.

$$f(x) = x^2 - 3x \text{ at } x = a + 1$$

Replace every  $x$  by  $(a + 1)$ , brackets included.

$$(a + 1)^2 - 3(a + 1) = a^2 - a - 2$$

Expand afterwards.

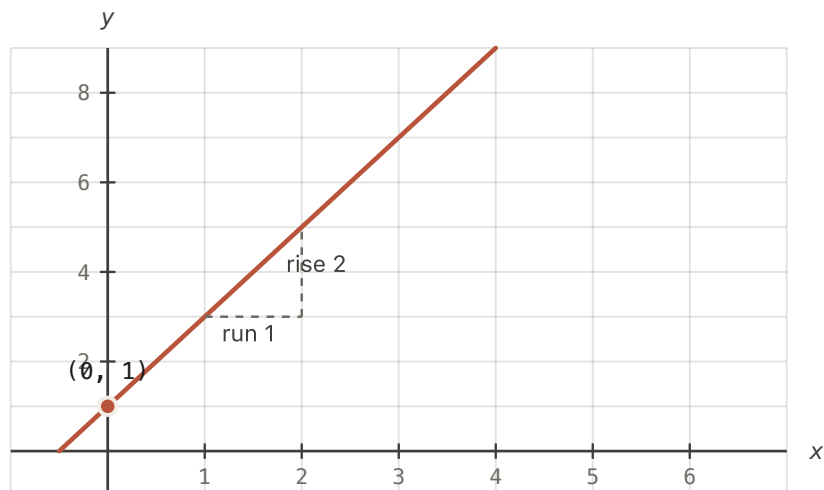
#### WATCH OUT

- $f(x)$  is one number, the output. It is not  $f$  times  $x$ .
- $f(a + 1)$  is not  $f(a) + 1$ .
- A vertical line meeting the graph twice means it is not a function: that input would have two outputs.

## Graphs, and the standard the course sets

### Lines

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$



A line through two points. The slope is the rise over the run, and it is the same wherever you measure it.

## The standard the course sets

### Sketching by hand

#### WATCH OUT

- Slope is  $\frac{\Delta y}{\Delta x}$ . Subtract the coordinates in the same order top and bottom.
- $x = k$  is vertical and has undefined slope, which is not slope zero.
- An unlabelled sketch earns no marks even when the curve is right.

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## 1.4 Domain and range

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► **Video** What is the domain of a function?

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► **Video** What is the range of a function?

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 **Exercise** Finding a domain · Determine the domain of functions

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 **Exercise** Reading both off a graph · Domain and range from graph

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### DOMAIN

Every input the rule accepts.

### RANGE

Every output that actually occurs. Usually smaller than you expect.

- 1 **Start by assuming every real number is allowed.** Then take away what is forbidden.
- 2 **A denominator may not be zero.** Set it to zero, solve, and exclude those values.
- 3 **An even root may not have a negative underneath.**  $\sqrt{x - 3}$  needs  $x \geq 3$ .
- 4 **A logarithm needs a strictly positive argument.**
- 5 **Solve each condition, then keep everything that survives all of them.**
- 6 **Write it whichever way you like.** An inequality, a set or an interval. All three are the same answer.
- 7 **From a graph, read both straight off the axes.** The domain is how far the curve reaches left and right, the range how far it reaches up and down.

**WORKED:**  $F(X) = X \div (X - 2)$

$$f(x) = \frac{x}{x - 2}$$

A quotient, so the denominator is the risk.

$$x - 2 = 0 \rightarrow x = 2$$

Find where the denominator vanishes.

**every real number except 2**

That is the domain.

**A SECOND:**  $F(X) = X \div (X^2 - X - 2)$

$$x^2 - x - 2 = 0$$

Same method, harder denominator.

$$(x - 2)(x + 1) = 0$$

Factor it.

**every real number except 2 and -1**

Two values removed, not one.

## When two things can go wrong at once

### Three ways to write it, and the range

#### WATCH OUT

- A root needs  $\geq 0$ ; a logarithm needs  $> 0$ , strictly.
- Simplifying first can hide a restriction. Take the domain from the formula as given.
- Context can shrink a domain the algebra allows: 7.3 employees is not an input.

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## 1.5 Six families of function

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▶ **Video**      Parabolic functions · Intro to parabolas

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▶ **Video**      Graphs of absolute value functions

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▶ **Video**      Graphs of rational functions

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 **Exercise**      Parabolas intro

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 **Exercise**      Case-defined functions · Evaluate piecewise functions

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### THE SIX FAMILIES

Constant, linear, quadratic, polynomial, rational, absolute value. Recognising the family from the formula tells you the shape before you draw anything.

- 1    **Look at the formula, not at the picture.** The form is what names the family.
- 2    **Linear: one x, first power.** A straight line.
- 3    **Quadratic: there is an  $x^2$ .** A parabola, opening upwards when  $a > 0$ .
- 4    **Rational: an x in a denominator.** So find the domain before anything else.
- 5    **Absolute value: distance from zero, so never negative.**

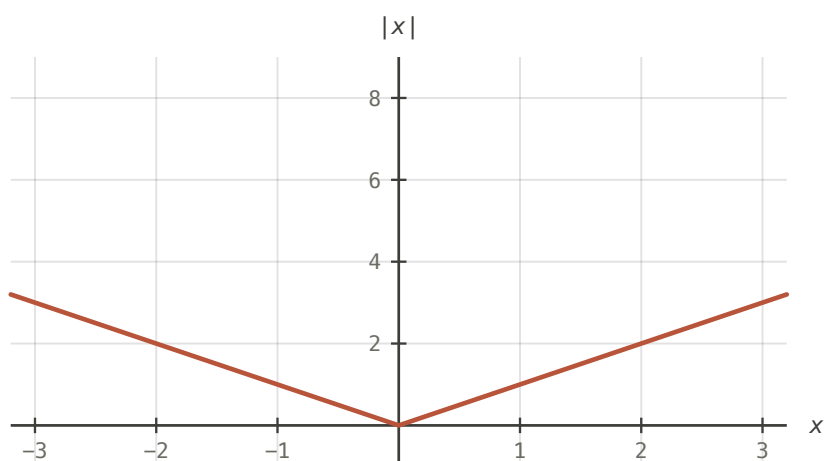
$$|x| = x \text{ if } x \geq 0, \quad -x \text{ if } x < 0$$

- 6    **Case-defined: several rules over several intervals.** Graph each piece only over its own interval, and check what happens at the joins.

| The six families |                                                                |                                    |
|------------------|----------------------------------------------------------------|------------------------------------|
| FAMILY           | DEFINITION                                                     | WHAT TO NOTICE                     |
| Constant         | $h(x) = c$                                                     | every input gives the same output  |
| Power            | $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+, x \rightarrow x^n$ | the variable is the base           |
| Parabolic        | $f(x) = ax^2 + bx + c$                                         | three constants, opens up or down  |
| Polynomial       | $a_n x^n + \dots + a_1 x + a_0$                                | $n$ is the degree                  |
| Rational         | $f(x) = \frac{P(x)}{Q(x)}$                                     | a quotient of polynomials          |
| Absolute value   | $ x $                                                          | two behaviours, one each side of 0 |

## Absolute value

$$|x| = x \text{ if } x \geq 0, \quad -x \text{ if } x < 0$$



The absolute value function. Two straight pieces meeting at the origin, one for each sign of  $x$ .

## Case-defined functions

### GRAPHED

$$f(x) = x \text{ for } 0 \leq x < 3$$

First piece: the line  $y = x$ .

$$f(x) = x - 1 \text{ for } 3 \leq x < 5$$

Second piece: the same slope, shifted down.

$$f(x) = 4 \text{ for } 5 \leq x < 7$$

Third piece: constant.

### WATCH OUT

- $x^2$  is a power function and  $2^x$  is exponential. The variable is in a different place.
- A case-defined function is one function, not three. It has one domain and one range.
- At a boundary, only one case applies. The inequality signs tell you which.

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## 1.6 Exponential functions

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- ▶ **Video**     What makes it exponential · Exponential vs. linear growth

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  - ▶ **Video**     Where e comes from ·  $e$  and compound interest

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  - ▶ **Video**      $e$  as a limit ·  $e$  as a limit
- 

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-  **Exercise**   Growth and decay in context · Exponential expressions word problems (numerical)

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  -  **Exercise**   The two shapes · Graphs of exponential growth
- 

### EXPONENTIAL FUNCTION

$f(x) = b^x$  with  $b > 0$  and  $b \neq 1$ . The variable is in the exponent, and that is what makes it exponential.

### GROWTH OR DECAY

Growth when  $b > 1$ , decay when  $b$  is between 0 and 1.

- 1    **Growing by  $p\%$  each period multiplies by the same number each period.**

$$\times \left(1 + \frac{p}{100}\right) \text{ each period}$$

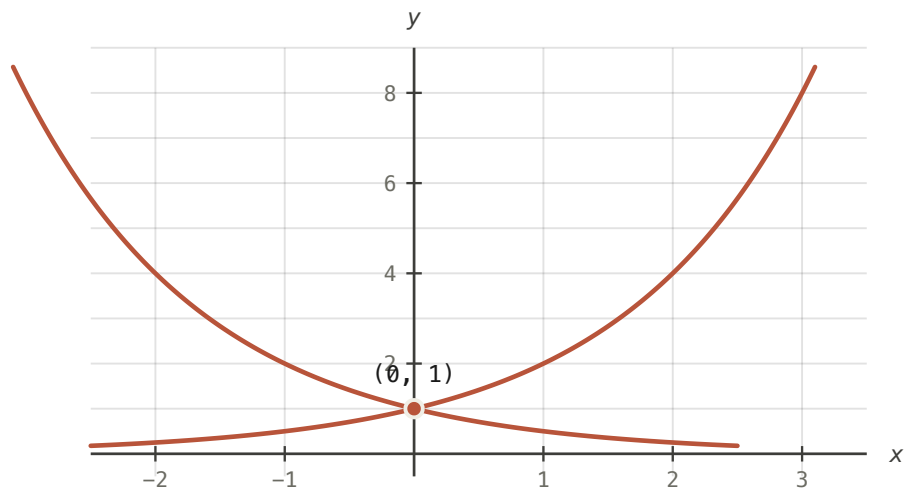
- 2    **Compound interest is the case you will meet most.**

$$S = P(1 + r)^n$$

$P$  the principal,  $r$  the rate per period,  $n$  the number of periods.

- 3    **Read the period off the question.** 8% a month and 8% a year are not the same number.
- 4     **$e$  is a limit, roughly 2.718.**  $\exp(x)$  and  $e^x$  are the same thing.

- 5 **A quantity that multiplies by a fixed factor each step is exponential.** One that adds a fixed amount each step is linear. That distinction is most of the exam questions.



Growth and decay.  $2^x$  doubles every step,  $0.5^x$  halves, and both pass through (0, 1).

## Compound interest, which is where it comes from

$$S = P(1 + r)^n$$

WORKED: 10 000 CHF AT 2% FOR THREE YEARS

$$S(3) = 10\,000 \times (1.02)^3$$

The multiplier is 1.02, not 0.02.

$$= 10\,612 \text{ CHF}$$

Rounded to the franc, to the franc.

## The number e

SUBSCRIBERS GROWING 8% A MONTH FROM 500

$$N(t) = 500 \cdot 1.08^t$$

Same shape as compound interest.

$$N(6) = 500 \cdot 1.08^6$$

Six months.

$$= 793 \text{ subscribers}$$

$$1.08^6 \approx 1.5869.$$

WATCH OUT

- The multiplier for a rise of  $r$  per cent is  $1 + \frac{r}{100}$ , not  $\frac{r}{100}$ .
- $b \neq 1$  is part of the definition:  $1^x$  is constant, not exponential.
- An exponential never reaches zero, so a decay question never answers "it becomes zero".

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## 1.7 Logarithmic functions

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- ▶ **Video**     What a logarithm is · Introduction to logarithms

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  - ▶ **Video**     The shape of the graph · Graphs of logarithmic functions

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  - ≡ **Article**    The definition, written out · Intro to logarithms
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- ✍ **Exercise**   Converting between the two forms · Relationship between exponentials & logarithms

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  - ✍ **Exercise**   Evaluating without a calculator · Evaluate logarithms

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  - ✍ **Exercise**   Graphing · Graphs of logarithmic functions
- 

### LOGARITHM

$$y = \log_b x \quad \text{if and only if} \quad b^y = x$$

The logarithm answers: to what power do I raise the base to get this number?

### THE TWO ON YOUR CALCULATOR

$\log$  is base 10.  $\ln$  is base  $e$ .

- 1    **To convert, ask what power the base is raised to.** Reading the statement the other way round is most of the work.
  
- 2    **Each undoes the other.**

$$\log_b(b^x) = x \quad b^{\log_b x} = x$$

- 3    **Without a calculator, write the argument as a power of the base.**  $\log_2 32$ : 32 is  $2^5$ , so the answer is 5.
  
- 4    **Only a strictly positive argument has a logarithm.** Check that before you do anything else.

- 5 To solve  $b$  to the power  $x$  equals  $k$ , take the log of both sides. The power rule then brings the  $x$  down where you can reach it.

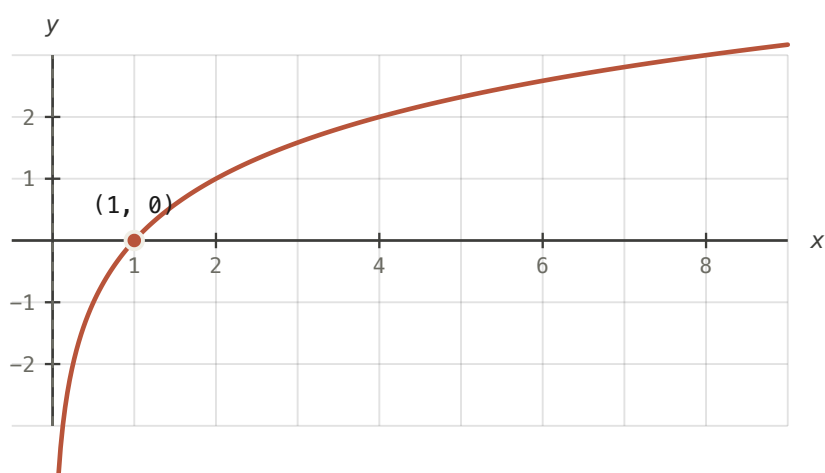
$$y = \log_b x \quad \text{if and only if} \quad b^y = x$$

$$\log_b(b^x) = x \quad b^{\log_b x} = x$$

## Converting between the two forms

| Conversions             |                             |
|-------------------------|-----------------------------|
| EXPONENTIAL FORM        | LOGARITHMIC FORM            |
| $5^2 = 25$              | $\log_5 25 = 2$             |
| $3^4 = 81$              | $\log_3 81 = 4$             |
| $10^0 = 1$              | $\log_{10} 1 = 0$           |
| $10^3 = 1000$           | $\log_{10} 1000 = 3$        |
| $64^{1/2} = 8$          | $\log_{64} 8 = \frac{1}{2}$ |
| $2^{-4} = \frac{1}{16}$ | $\log_2 \frac{1}{16} = -4$  |

## The two special logarithms



$y = \ln x$ . Defined only for  $x > 0$ , crossing at  $(1, 0)$ , falling without limit towards the axis.

#### WORKED, WITHOUT A CALCULATOR

$$\log 100 = 2$$

Because  $100 = 10^2$ .

$$\ln 1 = 0$$

Because  $e^0 = 1$ .

$$\log 0.1 = -1$$

Because  $0.1 = 10^{-1}$ .

$$\ln e^{-1} = -1$$

The two undo each other.

#### WATCH OUT

- $\log x$  needs  $x > 0$ , strictly. No power of a positive base gives zero or a negative.
- The graph falls without limit as  $x$  approaches 0. It never crosses the vertical axis.
- $\log(a + b)$  does not break up. There is no rule for the logarithm of a sum.

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## 1.8 Week 1 review quiz

Seven questions in the format of the Cyberlearn quiz: four answers and "none of the above". The wrong options are the answers you get from the usual slips, so read them before choosing.

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**Quantitative Methods I.** Notes I wrote while learning this. Free to use and to share.

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